

MEDILINK

Real-Time Hospital & Doctor Finder with E-Prescription System

A MINI-PROJECT REPORT

IT23431–MongoDB Essentials

Submitted by

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(An Autonomous Institution)



RAJALAKSHMI ENGINEERING COLLEGE
BONAFIDE CERTIFICATE

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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ABSTRACT

This project, titled “MEDILINK – Real-Time Hospital & Doctor Finder with E-Prescription System,” focuses on developing a smart and efficient web-based healthcare solution that connects patients with nearby hospitals and doctors in real time. The system is designed to simplify the process of finding medical assistance by providing accurate location-based results and instant access to healthcare services.

The application is developed using a modern web framework and integrates MongoDB as the primary database to handle large volumes of dynamic and unstructured healthcare data. MongoDB’s flexible schema design enables efficient storage and retrieval of information such as doctor profiles, hospital details, patient records, and electronic prescriptions.

The system offers key functionalities including real-time hospital and doctor search, user registration and authentication, appointment management, and a secure e-prescription module. Doctors can generate and share digital prescriptions, which are stored in the database and can be accessed by patients anytime. The platform ensures data consistency, scalability, and high performance through efficient backend operations.

This project demonstrates the practical application of MongoDB in building scalable healthcare systems and highlights the importance of digital transformation in improving accessibility, efficiency, and reliability in medical services. Overall, MEDILINK provides a user-friendly and technology-driven solution for modern healthcare management.

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

The rapid growth of technology has significantly transformed the healthcare sector, making medical services more accessible and efficient. However, many patients still face difficulties in finding nearby hospitals, available doctors, and managing prescriptions effectively. This project, *MEDILINK – Real-Time Hospital & Doctor Finder with E-Prescription System*, aims to bridge this gap by providing a centralized web-based platform that connects patients with healthcare providers in real time. The system allows users to search for hospitals and doctors based on location, book appointments, and receive digital prescriptions. By integrating MongoDB as the backend database, the application ensures efficient data storage, scalability, and flexibility in handling medical data.

1.2 SCOPE OF THE WORK

The scope of this project includes the development of a web-based application that enables users to locate nearby hospitals and doctors in real time. It covers functionalities such as user registration, login authentication, doctor and hospital search, appointment booking, and e-prescription management. The system is designed to be scalable and adaptable for future enhancements such as mobile application integration, real-time notifications, emergency services, and AI-based doctor recommendations. The project focuses on improving accessibility and reducing the time required for patients to receive medical assistance.

1.3 PROBLEM STATEMENT

In many areas, patients face challenges in quickly identifying nearby hospitals and available doctors, especially during emergencies. Traditional methods of searching for healthcare services are time-consuming and inefficient. Additionally, the management of medical prescriptions in paper format leads to issues such as loss, damage, and lack of accessibility. There is a need for a digital solution that provides real-time healthcare information and ensures secure storage and easy access to medical records and prescriptions

1.4 AIM AND OBJECTIVES OF THE PROJECT

Aim:

To develop a real-time web-based healthcare system that enables users to find hospitals and doctors efficiently and manage e-prescriptions digitally.

Objectives:

- To design a user-friendly interface for searching hospitals and doctors.
- To implement real-time location-based search functionality.
- To develop a secure user authentication system.
- To enable appointment booking between patients and doctors.
- To create an e-prescription system for digital medical records.
- To utilize MongoDB for efficient data storage and management.
- To ensure scalability and performance of the system.

CHAPTER 2

SYSTEM SPECIFICATIONS

2.1 Hardware Specification

- Processor : Intel Core i5
- RAM : Minimum 4 GB
- Hard Disk : 500 GB or above
- Input Devices : Keyboard and Mouse
- Output Devices : Monitor

2.2 Software Specification

- Operating System : Windows 10
- Frontend : HTML, CSS, JavaScript
- Backend : Python (Flask)
- Database : MongoDB
- Browser : Google Chrome / Microsoft Edge
- Tools : VS Code / Antigravity

CHAPTER 3

MODULE DESCRIPTION

3.1 User & Authentication Module

This module manages registration and login for patients, doctors, and hospital admins. It ensures secure access to the system with proper authentication and role-based control.

3.2 Hospital & Doctor Management Module

This module handles hospital details, doctor profiles, availability status, and specialization. Admins can manage resources such as beds, doctors on duty, and hospital information in real time.

3.3 Real-Time Search & Navigation Module

This module allows patients to find nearby hospitals and doctors using location-based services. It includes a live map and navigation support to help users reach the hospital quickly.

3.4 Appointment Management Module

This module enables patients to book appointments and hospitals to manage daily schedules. It tracks appointments and ensures smooth communication between patients and doctors.

3.5 Emergency & SOS Module

This is a critical module that allows users to send SOS alerts, share location, and request ambulances. Hospitals receive alerts in real time and can dispatch emergency services immediately.

3.6 E-Prescription & Medical Records Module

This module allows doctors to generate digital prescriptions and stores patient medical records securely. Patients can access their records anytime.

3.7 Pharmacy & Inventory Module

This module manages medicine availability and pharmacy approvals. It helps hospitals track inventory and allows patients to access prescribed medicines.

3.8 Alerts & Notification Module

This module provides real-time alerts such as emergency SOS, hospital updates, and system notifications to both patients and hospital staff.

3.9 Analytics & Dashboard Module

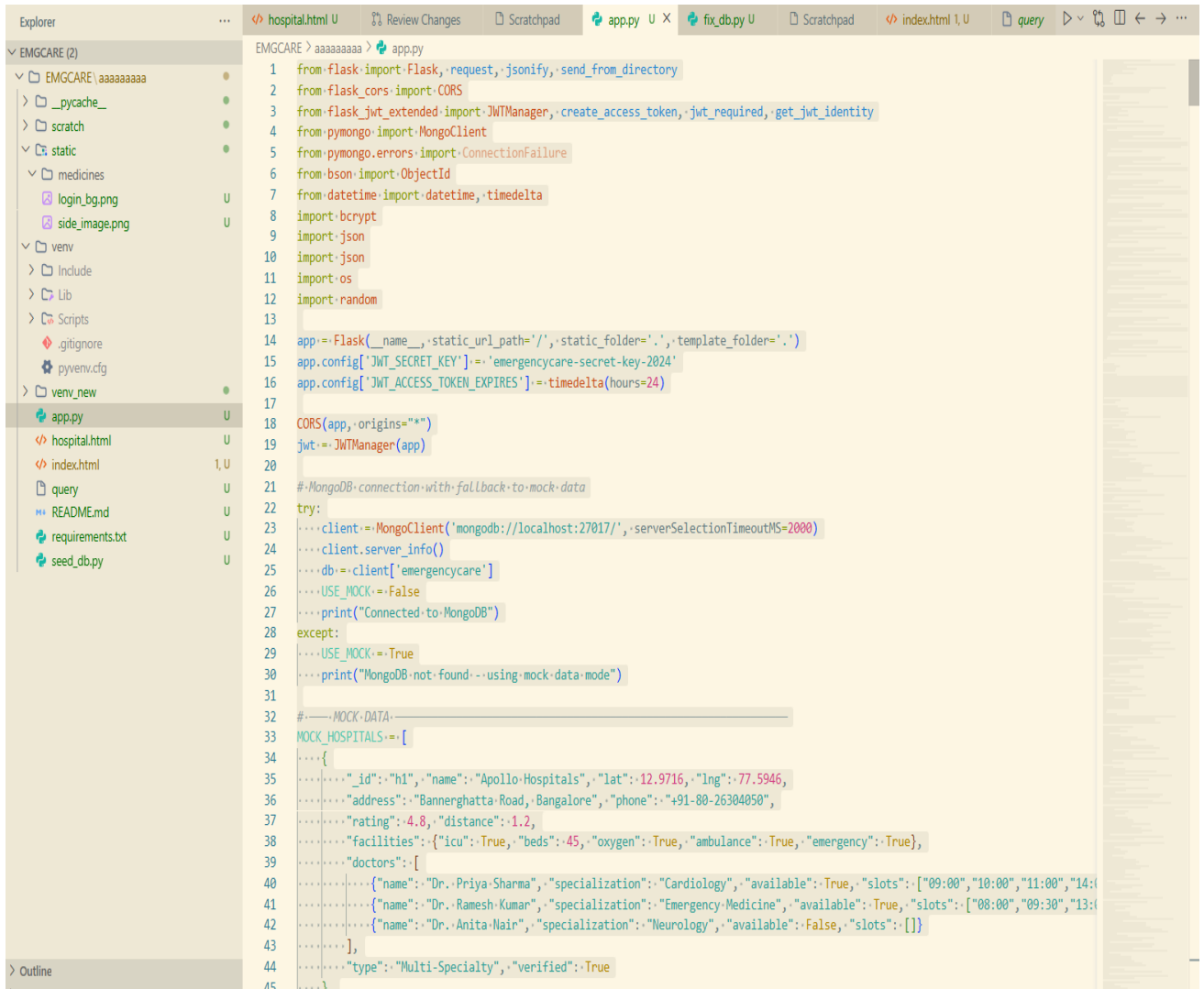
This module displays key data such as available beds, doctors on duty, appointments, and emergency cases through dashboards for quick monitoring and decision-making.

3.10 AI-Based Smart Assistance Module

This module uses Artificial Intelligence to provide intelligent support to users. It analyzes user inputs such as symptoms, location, and urgency to suggest nearby hospitals and suitable doctors. It also offers basic first-aid guidance and prioritizes emergency cases using smart recommendations. This improves decision-making speed and enhances the overall efficiency of the healthcare system.

CHAPTER 4

CODING



```
1 from flask import Flask, request, jsonify, send_from_directory
2 from flask_cors import CORS
3 from flask_jwt_extended import JWTManager, create_access_token, jwt_required, get_jwt_identity
4 from pymongo import MongoClient
5 from pymongo.errors import ConnectionFailure
6 from bson import ObjectId
7 from datetime import datetime, timedelta
8 import bcrypt
9 import json
10 import os
11 import random
12
13
14 app = Flask(__name__, static_url_path='', static_folder='.', template_folder='.')
15 app.config['JWT_SECRET_KEY'] = 'emgcare-secret-key-2024'
16 app.config['JWT_ACCESS_TOKEN_EXPIRES'] = timedelta(hours=24)
17
18 CORS(app, origins="*")
19 jwt = JWTManager(app)
20
21 # MongoDB connection with fallback to mock data
22 try:
23     client = MongoClient('mongodb://localhost:27017/', serverSelectionTimeoutMS=2000)
24     client.server_info()
25     db = client['emgcare']
26     USE_MOCK = False
27     print("Connected to MongoDB")
28 except:
29     USE_MOCK = True
30     print("MongoDB not found - using mock data mode")
31
32 # --- MOCK DATA ---
33 MOCK_HOSPITALS = [
34     {
35         "_id": "h1", "name": "Apollo Hospitals", "lat": 12.9716, "lng": 77.5946,
36         "address": "Bannerghatta Road, Bangalore", "phone": "+91-80-26304050",
37         "rating": 4.8, "distance": 1.2,
38         "facilities": {"icu": True, "beds": 45, "oxygen": True, "ambulance": True, "emergency": True},
39         "doctors": [
40             {"name": "Dr. Priya Sharma", "specialization": "Cardiology", "available": True, "slots": ["09:00", "10:00", "11:00", "14:00", "15:00", "16:00", "17:00", "18:00", "19:00", "20:00", "21:00", "22:00", "23:00", "24:00"]},
41             {"name": "Dr. Ramesh Kumar", "specialization": "Emergency Medicine", "available": True, "slots": ["08:00", "09:30", "13:00", "14:30", "18:00", "19:30", "22:00", "23:30", "24:00"]},
42             {"name": "Dr. Anita Nair", "specialization": "Neurology", "available": False, "slots": []}
43         ],
44         "type": "Multi-Specialty", "verified": True
45     }
46 ]
```

EMGCARE > aaaaaaaa > hospital.html > html > body > div.layout > main.main > div#page-overview.page.active

```
1 <!DOCTYPE html>
2 <html lang="en" data-theme="dark">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>EmergencyCare AI - Hospital Dashboard</title>
7 <link href="https://fonts.googleapis.com/css2?family=Syne:wght@400;600;700;800&family=DM+Sans:opsz,wght@9..40,300;9..40,400;9..40,500;9..40,600;9..40,700;9..40,800;9..40,900&display=block">
8 <style>
9 :root{--bg: #0a0e1a;--bg2: #0d1220;--surface: rgba(255,255,255,.04);--surface2: rgba(255,255,255,.07);--border: rgba(255,255,255,.07);}
10 *{margin:0;padding:0;box-sizing:border-box;}
11 body{font-family: 'DM Sans', sans-serif;background:var(--bg);color:var(--text);min-height:100vh;}
12 h1,h2,h3,h4{font-family: 'Syne', sans-serif;}
13 .layout{display:grid;grid-template-columns:250px 1fr;min-height:100vh;}
14 .sidebar{background:var(--bg2);border-right:1px solid var(--border);padding:24px 16px;display:flex;flex-direction:column;}
15 .logo{display:flex;align-items:center;gap:10px;margin-bottom:32px;padding:0 8px;}
16 .logo-icon{width:36px;height:36px;background:linear-gradient(135deg,var(--blue),var(--purple));border-radius:10px;display:flex;align-items:center;justify-content:center;}
17 .logo-h2{font-size:15px;font-weight:800;letter-spacing:-.3px;}
18 .logo-span{color:var(--blue);}
19 .nav-item{display:flex;align-items:center;gap:10px;padding:11px 12px;border-radius:10px;cursor:pointer;color:var(--text2);font-size:12px;}
20 .nav-item:hover{background:var(--surface);color:var(--text);}
21 .nav-item.active{background:rgba(77,159,255,.12);color:var(--blue);}
22 .main{padding:28px;overflow-y:auto;}
23 .topbar{display:flex;align-items:center;justify-content:space-between;margin-bottom:28px;}
24 .topbar-h2{font-size:22px;font-weight:800;}
25 .badge{padding:4px 10px;border-radius:20px;font-size:11px;font-weight:600;background:rgba(0,214,143,.12);color:var(--green);border:1px solid var(--border);}
26 .stats-row{display:grid;grid-template-columns:repeat(4,1fr);gap:14px;margin-bottom:24px;}
27 .stat-card{background:var(--surface);border:1px solid var(--border);border-radius:var(--radius);padding:18px;}
28 .stat-icon{font-size:22px;margin-bottom:8px;}
29 .stat-value{font-size:28px;font-weight:800;font-family:'Syne', sans-serif;line-height:1;}
30 .stat-label{font-size:12px;color:var(--text2);margin-top:4px;}
31 .two-col{display:grid;grid-template-columns:1fr 1fr;gap:20px;margin-bottom:24px;}
32 .card{background:var(--surface);border:1px solid var(--border);border-radius:var(--radius);overflow:hidden;}
33 .card-header{padding:16px 20px;border-bottom:1px solid var(--border);display:flex;justify-content:space-between;align-items:center;}
34 .card-title{font-family:'Syne', sans-serif;font-size:15px;font-weight:700;}
35 .card-body{padding:20px;}
36 .page{display:none;}
37 .page.active{display:block;animation:fadeIn .3s;}
38 @keyframes fadeIn{from{opacity:0;transform:translateY(6px);}to{opacity:1;transform:translateY(0)}}
39 .form-group{margin-bottom:16px;}
40 .form-group label{display:block;font-size:12px;font-weight:600;color:var(--text2);margin-bottom:6px;text-transform:uppercase;letter-spacing:.5px;}
41 .form-group input,.form-group select,.form-group textarea{width:100%;padding:12px 14px;background:var(--surface2);border:1px solid var(--border);border-radius:10px;}
42 .form-group select option{background:#1a2035;}
43 .btn-primary{padding:12px 24px;background:linear-gradient(135deg,var(--blue),#2563eb);border:none;border-radius:10px;color:#fff;font-weight:700;}
44 .btn-primary:hover{transform:translateY(-1px);}
45 .btn-danger{padding:8px 14px;background:rgba(255,59,92,.12);border:1px solid rgba(255,59,92,.3);border-radius:8px;color:var(--text2);font-weight:600;}
```

/* — NOISE TEXTURE OVERLAY — */

```
body::before {
  content: '';
  position: fixed;
  inset: 0;
  background-image: url("data:image/svg+xml,%3Csvg viewBox='0 0 256 256' xmlns='http://www.w3.org/2000/svg'%3E%3Cfilter id='noise'>%3CfeTurbulence type='fractalNoise' baseFrequency='0.001' numOctaves='4' stitchTiles='stitch'>%3C/feTurbulence%3C/filter%3E%3Crect width='100%' height='100%' fill='white' opacity='.4'>%3C/rect%3E%3C/svg>");
  pointer-events: none;
  z-index: 0;
  opacity: .4;
}
```

/* — AUTH SCREEN — */

```
#auth-screen {
  position: fixed;
  inset: 0;
  background: url('static/login_bg.png') center/cover no-repeat;
  display: flex;
  align-items: center;
  justify-content: center;
  z-index: 1000;
  transition: opacity .5s;
}
```

#auth-screen::before {

```
  content: '';
  position: absolute;
  inset: 0;
  background: radial-gradient(circle at center, rgba(77, 159, 255, 0.15) 0%, transparent 60%);
  animation: pulse-bg 6s infinite alternate;
  z-index: 0;
}
```

@keyframes pulse-bg {

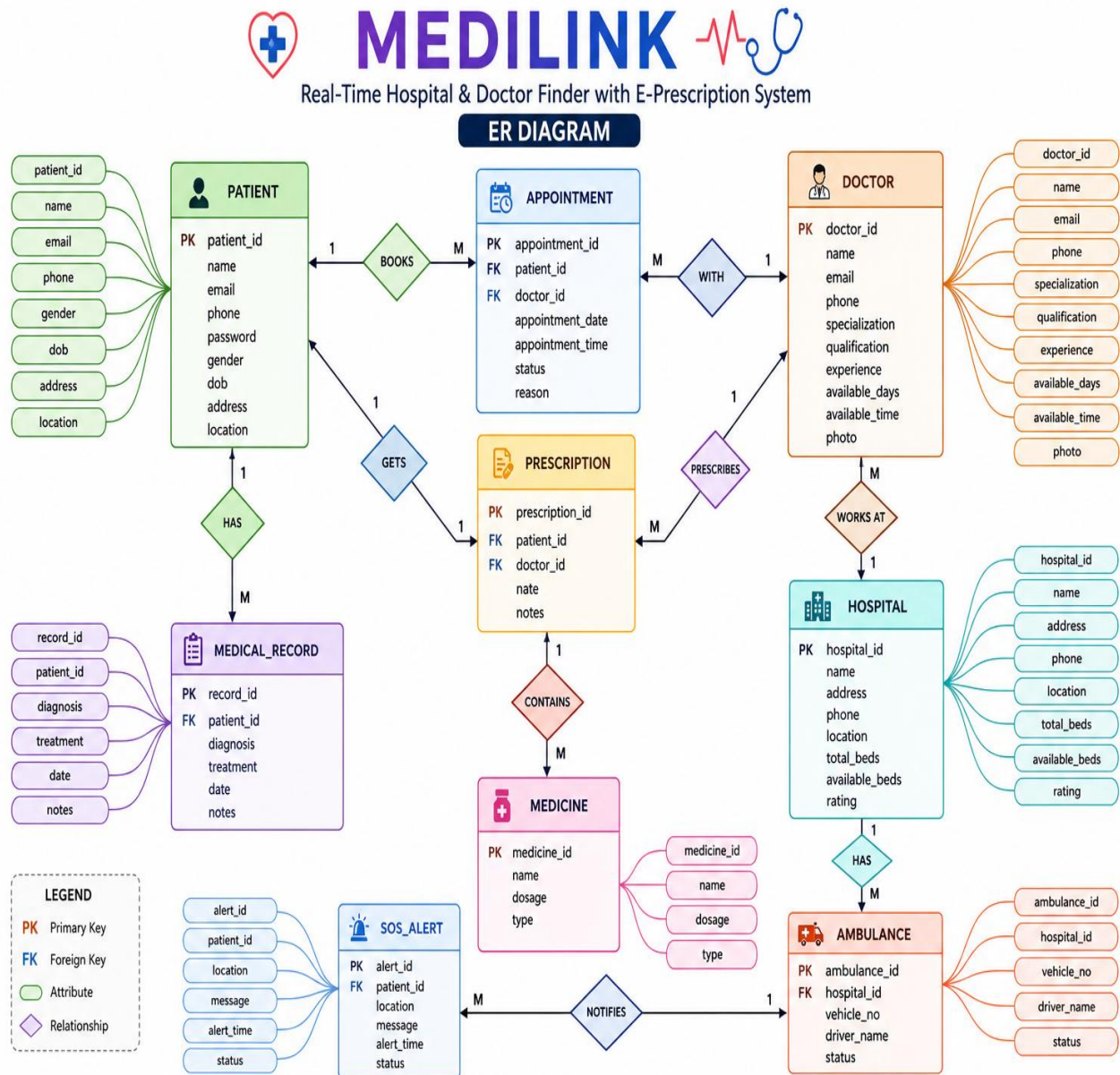
```
  0% { transform: scale(1); opacity: 0.5; }
  100% { transform: scale(1.2); opacity: 1; }
```

}

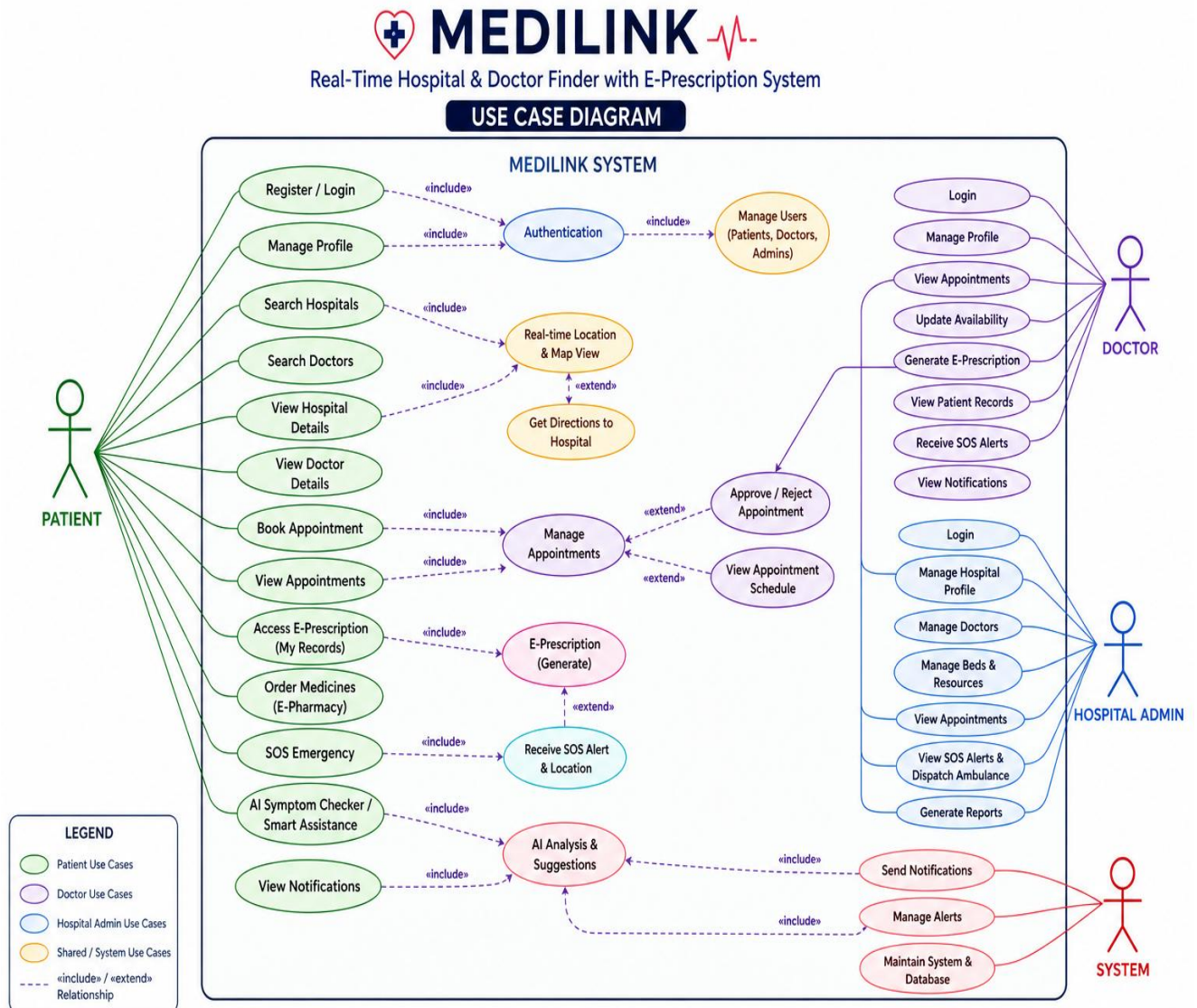
CHAPTER 5

DIAGRAMS

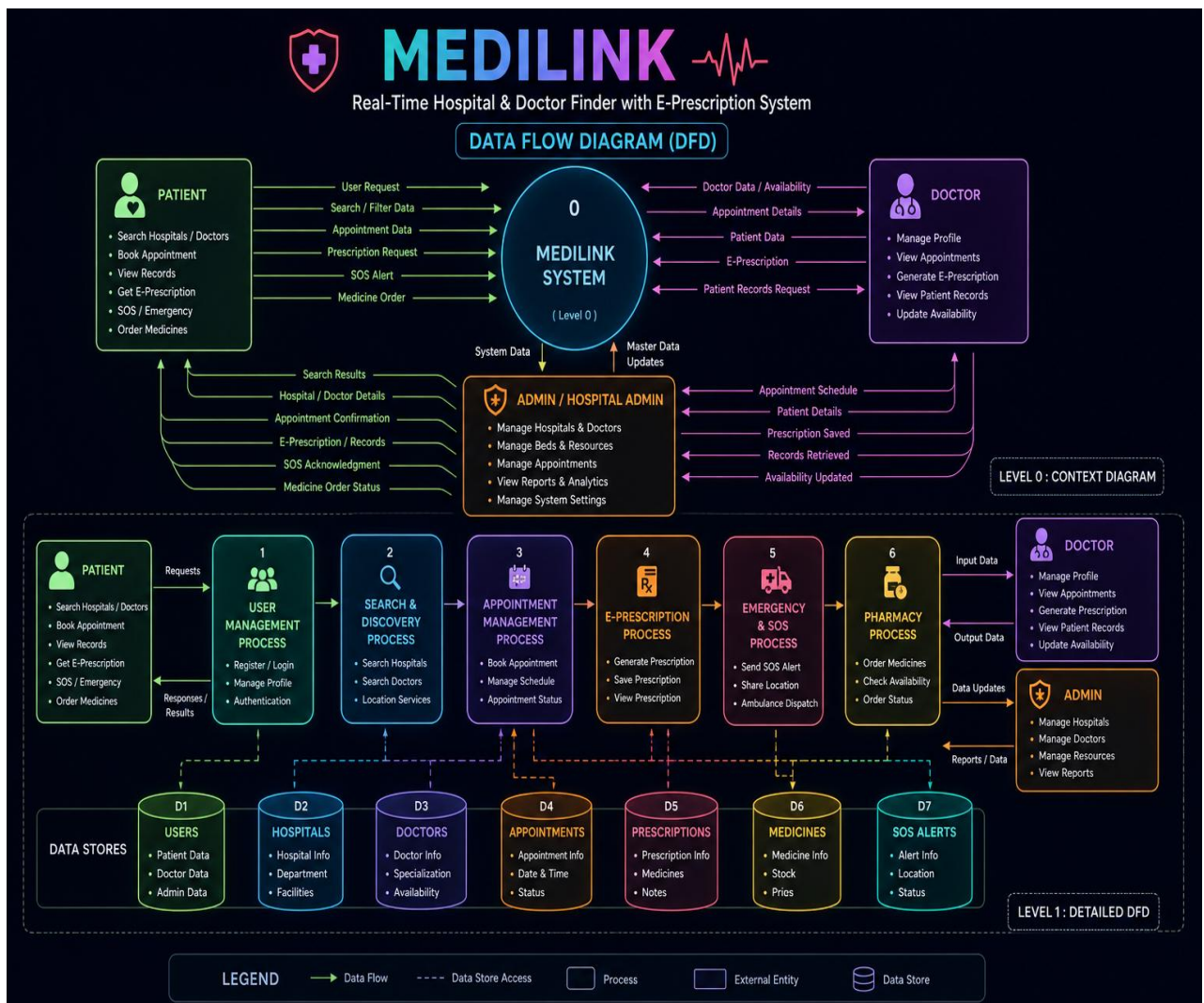
5.1 ER DIAGRAM



5.2 USE CASE DIAGRAM

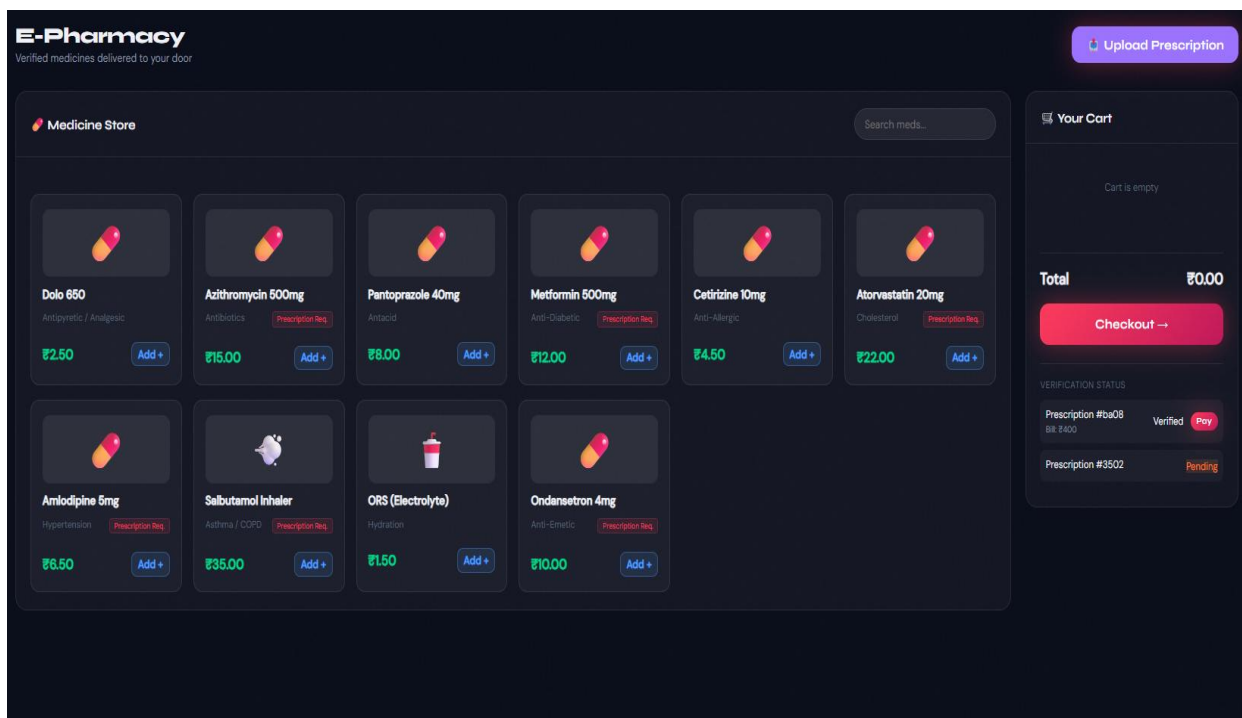
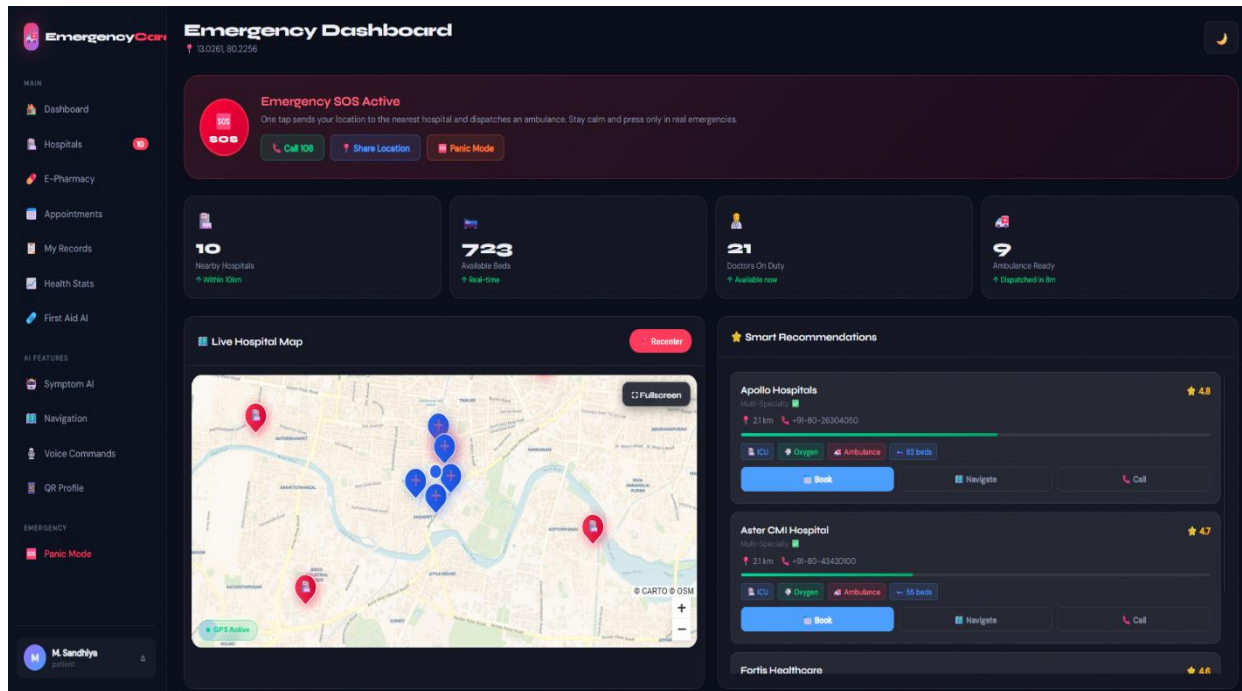


5.3 DATA FLOW DIAGRAM



CHAPTER 6

SCREENSHOTS



Appointments

Manage your bookings

[Book New](#)

8 MAY

Dr. Priya Sharma

Cardiology - Apollo Hospital - 0330

Heart Checkup

confirmed

23 MAY

Dr. Meera Iyer

Pediatrics - Fortis Healthcare - 0330

Vaccination

pending

Reschedule

3 MAY

Dr. Lakshmi Rao

General Surgery - Narayana Health - 0330

Lap HERN

pending

Reschedule

13 MAY

Dr. Pooja Singh

Gynecology - Manipal Hospital - 0930

Pregnancy scan

pending

Reschedule

17 JUN

Dr. prathheeba

General Doctor - Apollo Hospital - 0330

Neck pain checkup

pending

Reschedule

SOS

PANIC MODE ACTIVATED

2

Calling ambulance & alerting contacts in...

CANCEL

AI Symptom Checker

Get urgency assessment instantly

Select Symptoms

Chest Pain

Severe Headache

Abdominal Pain

Allergic Reaction

Stroke Signs

Eye Pain

Difficulty Breathing

Vomiting

Back Pain

Broken Bone

Heart Palpitations

High Fever

Dizziness

Seizure

Unconscious

Severe Bleeding

ADDITIONAL NOTES

Describe any additional symptoms...

Analyze Symptoms →

CRITICAL URGENCY

CALL 108 IMMEDIATELY. This is a life-threatening emergency!

Recommended: Cardiology

Find Hospital →

Smart Hospital Match

Apollo Hospitals

Multi-Specialty

12 km

+91-80-26304050

ICU

Oxygen

Ambulance

45 beds

Book

Navigate

Call

NIMHANS

Psychiatric Specialty

4.5 km

+91-80-26995000

ICU

Oxygen

Ambulance

200 beds

Book

Navigate

Call

Fortis Healthcare

Multi-Specialty

2.8 km

+91-80-66214444

ICU

Oxygen

Ambulance

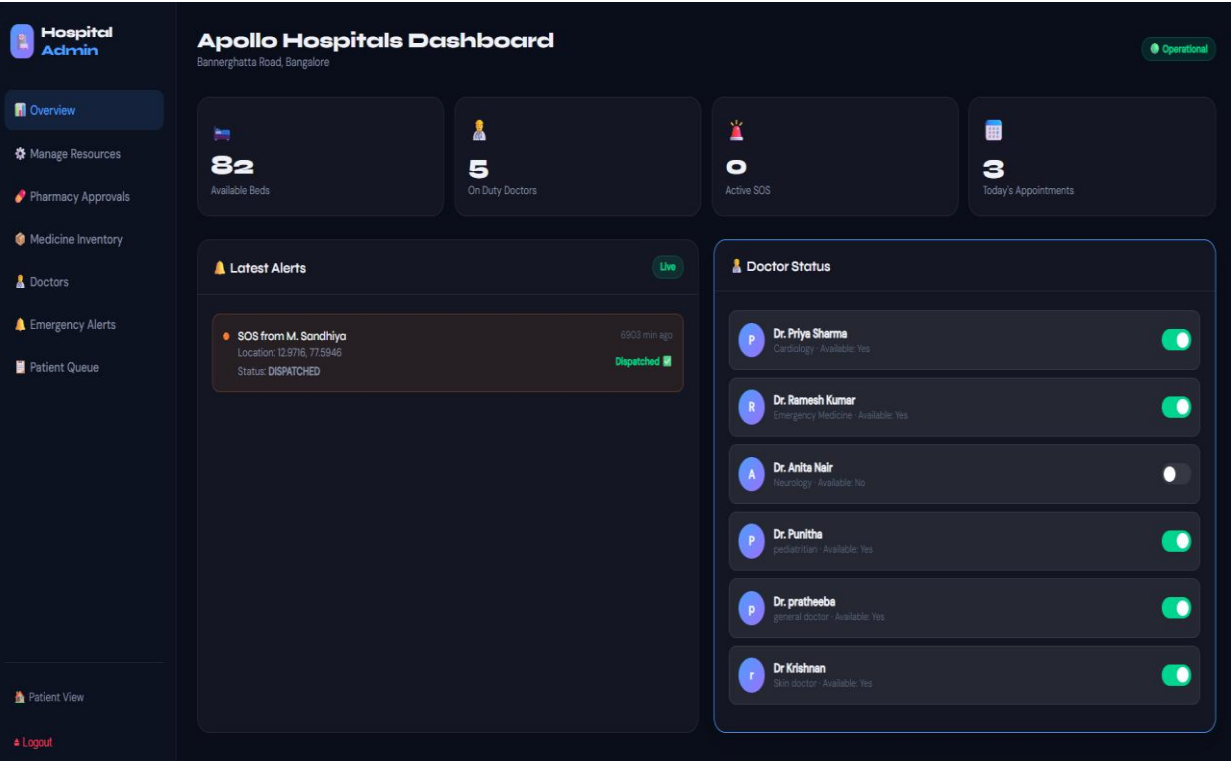
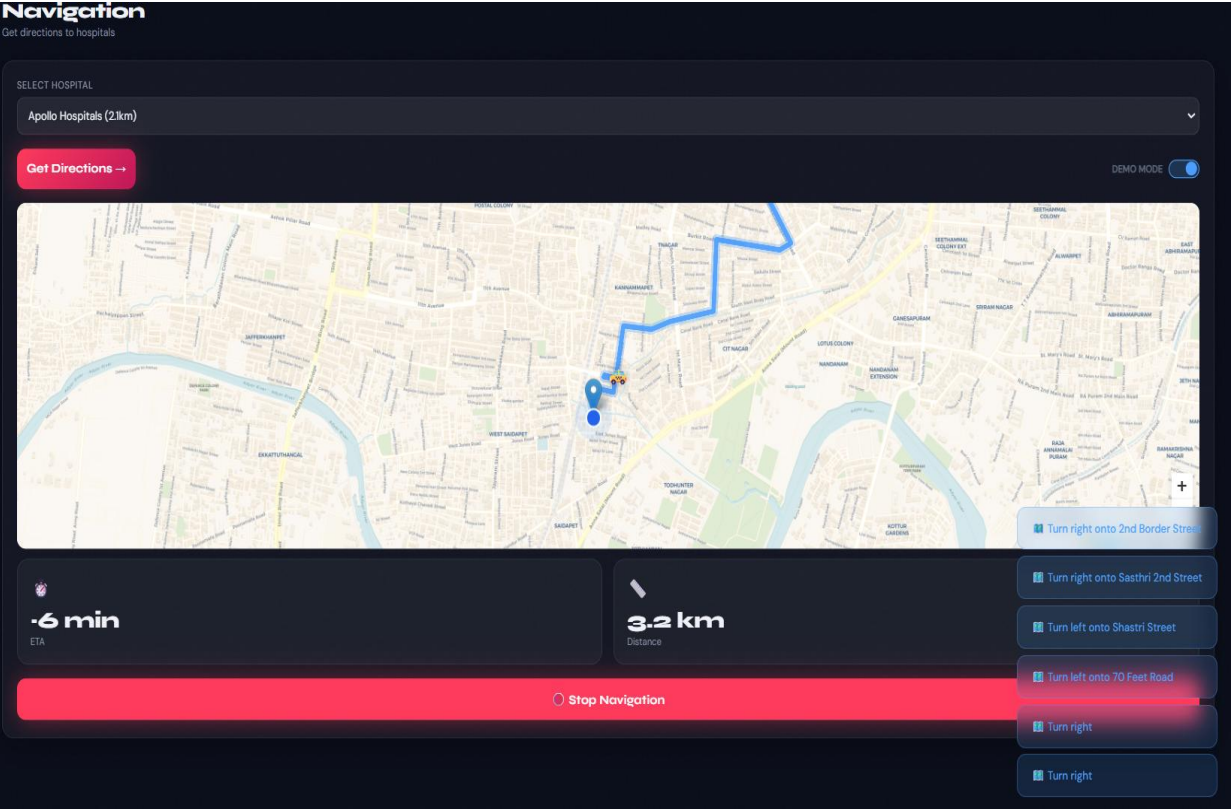
28 beds

Book

Navigate

Call

18



CONCLUSION

The project “MEDILINK – Real-Time Hospital & Doctor Finder with E-Prescription System” successfully demonstrates the development of a modern healthcare solution that improves accessibility and efficiency in medical services. The system enables users to locate nearby hospitals and doctors in real time, book appointments, and manage digital prescriptions effectively. By integrating MongoDB, the application ensures flexible data storage, scalability, and efficient handling of healthcare information.

The inclusion of features such as emergency SOS alerts, real-time availability tracking, and AI-based assistance enhances the usability and responsiveness of the system. Overall, the project highlights the importance of digital transformation in healthcare and provides a reliable platform that can be further expanded into a full-scale real-world application.

REFERENCES

- MongoDB Official Documentation – <https://www.mongodb.com/docs/>
- Flask Documentation – <https://flask.palletsprojects.com/>
- W3Schools Web Development Tutorials – <https://www.w3schools.com/>
- MDN Web Docs (HTML, CSS, JavaScript) – <https://developer.mozilla.org/>
- Google Maps API Documentation – <https://developers.google.com/maps>